

1. The game will generate grade-appropriate math problems aligned with 6-8th grade standards
 - 1.1. The game will show one of 4 different types of problems
 - 1.1.1. The game will create linear equation problems ($ax + b = c$ format)
 - 1.1.2. The game will create systems of equations problems
 - 1.1.3. The game will create geometry problems (area, perimeter, volume)
 - 1.1.4. The game will create percentage and ratio problems
 - 1.2. The game will present math problems through an interactive UI overlay
 - 1.3. The game will provide immediate feedback on whether the student answered the question correctly
2. The game will contain resources
 - 2.1. The game will contain these resources types: wood, scrap, and food
 - 2.2. Resources are stored in an inventory location which can be accessed at anytime by the player
3. The game will contain four explorable islands
 - 3.1. The first island will contain a shop where players can purchase and upgrade equipment
 - 3.1.1. Equipment will be purchased with wood and scrap
 - 3.1.2. The player can purchase weapons, armor, and upgrades
 - 3.2. Each island will contain a set amount of zombies that need to be defeated before unlocking the next island
 - 3.2.1. Players will engage with zombies by running into them or by zombies running into them
 - 3.2.2. The player must answer timed math questions while engaged with the zombie
 - 3.2.2.1. If the player answers incorrectly or not in time, the zombie will deal damage to the player
 - 3.2.2.2. If the player answers correctly, the player will deal damage to the zombie
 - 3.2.3. Defeating a zombie will award resources to the player
 - 3.2.4. After the first island, defeating every zombie will spawn a chest in the center of the island that contains resources
 - 3.2.5. To complete the game, the player must defeat every zombie on every island
 - 3.3. Each island will contain a dock used to travel to other islands
 - 3.3.1. Islands 2 - 4 will be locked off in the beginning
 - 3.3.2. The player must correctly answer a slope problem before travelling to a new island
 - 3.3.2.1. If the player incorrectly answers the problem, they will have to fight a shark

- 3.3.2.1.1. The player must answers timed math questions while engaged with the shark
 - 3.3.2.1.1.1. If the player answers incorrectly or not in time, the shark will deal damage to the raft
 - 3.3.2.1.1.2. If the player answers correctly, the player will deal damage to the shark
 - 3.4. The game will trigger timed math challenges when the player enters a new island
 - 3.4.1. The player will have the option to decline the challenge with no penalty
 - 3.4.2. The player will be awarded resources if the challenge is completed in time
 - 3.4.3. The player will lose resources if the challenge is not completed in time
- 4. The game will end if player health or raft health reach 0
 - 4.1. Food can be used to restore player health
 - 4.2. Wood can be used to restore raft health
- 5. The game will have a day/night cycle
 - 5.1. The cycle corresponds to an in-game clock loop
 - 5.2. Map visibility will decrease at night
 - 5.3. Zombies and sharks deal more damage at night
- 6. The game will display all necessary information only during the day
 - 6.1. The game will display all resource quantities and equipment
 - 6.2. The game will show the current day and time
 - 6.2.1. Sharks and zombies deal more damage as the day counter goes up
 - 6.3. The game will show math problem text, input field, and allow the player to submit their answer
- 7. The game will contain different types of equipment
 - 7.1. Equipment is stored in an inventory location which can be accessed at anytime by the player
 - 7.2. The game will contain weapons
 - 7.2.1. Better weapons will deal more damage
 - 7.3. The game will contain armor
 - 7.3.1. Better armor will reduce damage taken
 - 7.4. The game will contain upgrades
 - 7.4.1. Upgrades will consist of night vision goggles
 - 7.4.1.1. Night vision goggles will improve the game world and UI visibility at night allowing the user to see as if it was daytime
 - 7.4.2. Upgrades will last permanently once obtained
 - 7.5. Players will start the game with only a sword
- 8. The game will play background music while the player explores the world

